

Vector Databases & Embeddings

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How to use this file: This is a deep, mechanism-level reference for FAANG SDE/ML interviews on vector search, embeddings, ANN indexes, and RAG retrieval. Read top to bottom once to build the mental model; then use the **Revision Cheat Sheet** and **Common Interview Questions** the night before. Every section pairs an explanation with real code, concrete numbers, and a trade-off. When an interviewer says "design semantic search over 100M docs," the **Worked Example** section is your script.

1 Overview --- What It Is

A **vector database** is a system that stores high-dimensional numeric vectors (**embeddings**) and answers the query "**which stored vectors are closest to this query vector?**" in milliseconds, even across hundreds of millions of vectors.

An **embedding** is a dense array of floating-point numbers — say 768 of them — that a neural network produces to represent the *meaning* of a piece of content (a sentence, a paragraph, an image, a product). Two pieces of content that mean similar things land near each other in this 768-dimensional space; unrelated content lands far apart. The vector database makes "find the nearest meanings" fast.

Layer	What it does	Examples
Embedding model	Turns text/image into a dense vector	text-embedding-3-small, bge-large, e5, CLIP
Distance metric	Defines "closeness" between vectors	cosine, dot product, L2 (Euclidean)
ANN index	Finds approximate nearest neighbors fast	HNSW, IVF, IVF-PQ, ScaNN, Annoy, LSH
Vector store	Persists vectors + metadata, serves queries	Pinecone, Weaviate, Milvus, Qdrant, pgvector
Retrieval pipeline	Chunk → embed → upsert → retrieve → rerank	RAG, semantic search, recommendations
Filtering	Restrict search by metadata (date, tenant, ACL)	"only docs from tenant X, after 2024"

Key insight: A vector database is not a replacement for Postgres. It is a specialized index — the moral equivalent of a B-tree, but for *semantic proximity* in high-dimensional space instead of *sorted order* on a scalar key. The hard engineering problem it solves is that **exact nearest-neighbor search in high dimensions is brutally slow**, so it trades a tiny amount of accuracy (recall) for orders-of-magnitude speedups.

```
"How do I reset my password?" -embed→ [0.12, -0.43, 0.87, ..., 0.31] (768 floats)
"Steps to recover account access" -embed→ [0.11, -0.41, 0.85, ..., 0.29] ← very close
"What is photosynthesis?" -embed→ [-0.67, 0.82, -0.34, ..., 0.91] ← far away
```

```
Vector DB job: given the first vector, return the second (and not the third)
                from among 100,000,000 stored vectors in < 50 ms.
```

2 Why It Exists

The core problem: meaning is not keywords.

Traditional search (inverted index + BM25, the engine behind Elasticsearch and classic Google) matches **tokens**. If you search "reset password" it finds documents containing the literal words "reset" and "password." This breaks in three ways:

1. **Synonymy** — "recover account access" means the same thing but shares zero keywords with "reset password." Keyword search misses it entirely.
2. **Polysemy** — "Apple stock" (fruit inventory vs. AAPL shares) — the same token, different meanings. Keyword search can't tell them apart.
3. **Paraphrase / intent** — "my laptop won't turn on" should match a doc titled "troubleshooting power issues on portable computers." No shared tokens.

Embeddings solve this because the model maps *meaning* to *geometry*. "Reset password" and "recover account access" produce nearly identical vectors even though they share no words.

The second core problem: exact nearest-neighbor search is too slow.

Once you have vectors, you need to find the nearest ones. The naive approach — compute the distance from the query to every stored vector and sort — is **exact k-Nearest-Neighbor (kNN)**, and it costs $O(N \cdot d)$ per query.

```

1 N = 100,000,000 documents
2 d = 768 dimensions
3 Cost per query = N × d = 76.8 billion multiply-adds
4
5 At ~10 GFLOP/s effective for a brute-force dot-product loop,
6 that's ~7.6 seconds per query. Per query. Unusable.

```

You might think: "use a tree, like a k-d tree, the way we'd do 2D nearest-neighbor." That fails because of the **curse of dimensionality**:

2.1 Why k-d trees fail in high dimensions

A k-d tree partitions space by axis-aligned splits and prunes branches that can't contain a closer point. In 2D or 3D this prunes ~99% of the tree and gives $O(\log N)$ search. But as dimensions grow:

- ▶ **Volume concentrates in the corners.** The fraction of a hypercube's volume near the surface goes to 1. In 768 dimensions, essentially all points are "near the boundary," so the pruning bound rarely lets you skip a branch.
- ▶ **Distances concentrate.** The ratio of the farthest to the nearest point distance approaches 1 as $d \rightarrow \infty$. When the nearest and farthest neighbors are nearly equidistant, "nearest neighbor" is barely meaningful and pruning is impossible.
- ▶ **Branching factor explosion.** To prune effectively in d dimensions you'd need $\sim 2^d$ cells; for $d=768$ that's astronomically more than any dataset.

Exact NN search method	Works well up to dimension
k-d tree / ball tree	~10-20
Cover tree	~30-50
Brute force ($O(N \cdot d)$)	any d , but $O(N)$ per query — too slow at scale
ANN (HNSW/IVF/PQ)	hundreds to thousands ← the only practical answer

This is exactly why **Approximate** Nearest Neighbor (ANN) exists: give up the guarantee

of finding *the* exact nearest neighbor, accept finding *almost always the right ones* (e.g., 95–99% recall), and get a 100–1000× speedup. The “approximate” is the whole point.

Before vector databases: semantic search required hand-built NLP — TF-IDF, latent semantic analysis (LSA/SVD), query expansion with synonym dictionaries, and entity linking. Each was brittle and shallow.

After embeddings + ANN: a single embedding model plus an ANN index gives deep semantic matching out of the box, and it generalizes to images, audio, and code with the same machinery.

3 Why FAANG Cares

Google: ANN is foundational. **ScaNN** (Scalable Nearest Neighbors) is Google's own library, used in Search and YouTube recommendations. Vertex AI Matching Engine serves billion-vector indexes. Embedding-based retrieval is how Google does semantic ranking, “related searches,” and RAG grounding for Gemini. Expect questions on ScaNN's anisotropic quantization and on grounding LLMs.

Meta: Built **FAISS** (Facebook AI Similarity Search) — the most-used ANN library in the world. Meta runs embedding retrieval for feed ranking, ads retrieval (find candidate ads from billions), Reels recommendations, and content moderation (find near-duplicate / similar harmful content). “Two-tower” retrieval models feeding FAISS are a core interview topic.

Amazon: Product search and recommendations (“customers who bought this”) are embedding retrieval at planetary scale. OpenSearch (their Elasticsearch fork) ships kNN/HNSW. Amazon Bedrock Knowledge Bases is managed RAG. Cost-per-query and memory footprint directly hit AWS margins, so quantization (PQ) and sharding come up.

Microsoft: Azure AI Search has native vector + hybrid search; it underpins Microsoft 365 Copilot's retrieval over your org's documents. They care about hybrid (dense + BM25) and ACL-aware filtered search (you must only retrieve docs you're allowed to see).

Apple: On-device semantic search (Spotlight, Photos “find pictures of dogs”), privacy-first, tight memory budgets — making PQ/compression and small embedding models matter.

Netflix / Spotify / Pinterest: Recommendations are retrieval. Spotify built **Annoy** (Approximate Nearest Neighbors Oh Yeah) for music recs. Pinterest's PinSage/visual search is embedding retrieval over billions of pins.

Startups (Perplexity, Glean, Harvey, Notion AI, Cursor): The product *is* retrieval quality. Every one of them ships RAG; chunking, hybrid search, and reranking are daily decisions.

Why interviewers ask: RAG is now standard in every AI product, so retrieval quality is a core competency. The recall/latency/memory trade-off is a clean systems-design question with real numbers. And it connects ML (embeddings, training objectives) with systems (indexes, sharding) — a great signal for senior generalists.

4 Core Concepts

4.1 Semantic search vs. lexical search

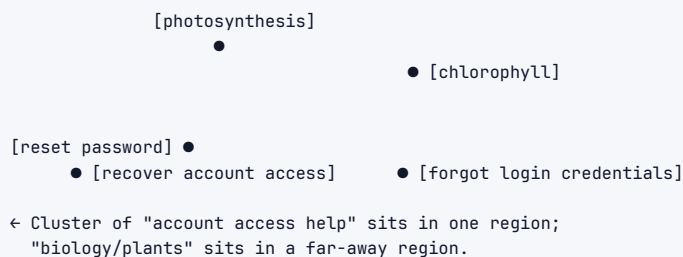
	Lexical (BM25 / inverted index)	Semantic (embeddings + ANN)
Matches on	Exact tokens / stems	Meaning / intent
Strength	Rare terms, IDs, names, acronyms, code	Paraphrase, synonyms, concepts
Weakness	Synonyms, paraphrase	Exact strings (“error code E-1042”), out-of-domain jargon

	Lexical (BM25 / inverted index)	Semantic (embeddings + ANN)
Index	Posting lists (sparse)	Dense vectors in ANN index
Scoring	TF-IDF / BM25	Cosine / dot product distance
Best in practice	—	Hybrid: use both, fuse the scores

The professional answer is almost never "semantic only." It's **hybrid search** — semantic for recall on meaning, lexical for precision on exact terms — fused together (covered later).

4.2 The vector space

After embedding, every document is a point in $\mathbb{R}^d$. The geometry encodes semantics:



Three operations matter:

- ▶ **Nearest-neighbor query** — given a query point, return the k closest stored points (search/RAG).
- ▶ **Clustering** — group points into regions (used internally by IVF; also for dedup, topic discovery).
- ▶ **Filtered NN** — nearest neighbors *subject to* a metadata predicate ("only points where tenant=X").

4.3 Exact kNN cost recap

```

1 import numpy as np
2
3 def exact_knn(query, vectors, k):
4     # vectors: (N, d) matrix, query: (d,)
5     # 0(N*d) dot products + 0(N log k) selection
6     sims = vectors @ query          # N*d multiply-adds
7     idx = np.argpartition(-sims, k)[:k]
8     return idx[np.argsort(-sims[idx])]
```

At $N=100M$, $d=768$ this is ~ 7.6 s/query (shown above). ANN indexes cut this to single-digit milliseconds by *not* looking at most vectors — they navigate a graph (HNSW) or only probe a few clusters (IVF).

Interview takeaway: Brute force is fine up to $\sim$ tens of thousands of vectors (a few ms). Past $\sim 100K$ – $1M$ you need an ANN index. State this threshold explicitly.

5 Embeddings

5.1 What a dense embedding is

A dense embedding is a fixed-length vector of real numbers where **every dimension carries (distributed) information** — unlike a sparse one-hot or bag-of-words vector where most entries are zero. "Dense" means all d entries are meaningful floats; "embedding" means the vector lives in a learned space where geometric closeness $\approx$ semantic similarity.

```
Sparse (bag-of-words, vocab=50,000): [0,0,...,1,...,0,1,0,...,0] ← 50,000 dims, mostly 0
Dense (embedding):                  [0.12, -0.43, 0.87, ..., 0.31] ← 768 dims, all informative
```

No single dimension means "is about cats." Meaning is **distributed** across all dimensions — direction in the space encodes concepts.

5.2 How models learn embeddings --- the training objective

Embeddings are learned, not designed. The dominant objective is **contrastive learning**: pull semantically-related pairs together and push unrelated pairs apart.

Contrastive / InfoNCE loss (the core idea):

For an anchor a , a positive p (something that should be similar), and negatives n_i (things that should be dissimilar), minimize:

$$\text{loss} = -\log\left(\frac{\exp(\text{sim}(a, p) / \tau)}{\sum_i \exp(\text{sim}(a, n_i) / \tau)} \right)$$

Maximizing the numerator pulls a and p together; the denominator pushes a away from all negatives. τ is a temperature. **In-batch negatives** are the trick that makes this cheap: within a batch of B (anchor, positive) pairs, every other pair's positive serves as a negative, giving $B-1$ negatives for free.

Self-supervised pretraining generates the pairs without human labels:

- › **Inverse cloze / span corruption** — a sentence and its surrounding paragraph are a positive pair.
- › **Dropout-as-augmentation (SimCSE)** — feed the *same* sentence through the encoder twice with different dropout masks; the two outputs are a positive pair. Brilliantly simple and strong.
- › **Title-body, question-answer, query-clicked-document** mined from logs are natural positives.

Sentence-transformers & bi-encoders: A **bi-encoder** encodes the query and the document *independently* into vectors, then compares with a cheap dot product. This is what makes retrieval scalable: you embed all N documents *once, offline*, store the vectors, and at query time only embed the query (one forward pass) and do ANN search. Contrast with a **cross-encoder** (used for reranking) which feeds [query, doc] *together* through attention — far more accurate but $O(N)$ forward passes, impossible at retrieval scale.

```
1 from sentence_transformers import SentenceTransformer
2
3 model = SentenceTransformer("BAAI/bge-base-en-v1.5") # bi-encoder, 768-dim
4
5 # Offline: embed the whole corpus once
6 doc_vecs = model.encode(documents, batch_size=256, normalize_embeddings=True)
7
8 # Online: embed only the query (one forward pass), then ANN search
9 q = model.encode("how do I reset my password", normalize_embeddings=True)
10 # similarity = dot product because vectors are unit-normalized
```

5.3 Dimensionality --- 384 vs 768 vs 1536 vs 3072

Dims	Typical models	Pros	Cons
384	all-MiniLM-L6-v2, bge-small	Tiny, fast, 4× less RAM than 1536	Lower ceiling on quality
768	bge-base, e5-base, nomic-embed	Sweet spot quality/cost	Standard default
1024	bge-large, cohere-embed-v3	Strong quality	More RAM/compute
1536	OpenAI text-embedding-3-small, ada-002	Excellent, managed API	2× RAM vs 768
3072	OpenAI text-embedding-3-large	Best OpenAI quality	4× RAM vs 768; often overkill

The trade-off is concrete. Memory and compute scale linearly with d:

```
100M vectors, float32 (4 bytes):
d=384 → 100M × 384 × 4 B = 153.6 GB
d=768 → 100M × 768 × 4 B = 307.2 GB
d=1536 → 100M × 1536 × 4 B = 614.4 GB
d=3072 → 100M × 3072 × 4 B = 1.23 TB
```

Doubling dimensions doubles RAM, doubles distance-computation cost, and doubles network/storage. Higher d usually improves recall — but with diminishing returns. **Matryoshka embeddings** (MRE, used by OpenAI v3 and nomic) train so that the *first* k dimensions are themselves a usable embedding — you can truncate 1536→512 and keep ~90%+ of quality, getting a runtime knob for the cost/quality trade-off.

5.4 Normalization --- unit vectors

Most text embeddings are **L2-normalized** to unit length ($\|v\| = 1$). This matters because:

- After normalization, **dot product = cosine similarity** (proof below), so you can use the faster inner product metric in the index and still get cosine ranking.
- It removes magnitude as a confound — you compare *direction* (meaning) only, not length.

```
1 v = v / np.linalg.norm(v) # now ||v|| = 1; dot(a,b) = cosine(a,b)
```

If your model emits normalized vectors, configure the index for **inner product (IP)**, not L2 — it's the same ranking and slightly cheaper.

5.5 Choosing / fine-tuning a domain model

Off-the-shelf models are trained on web/general text. For specialized domains (legal, biomedical, code, your internal jargon) they under-perform because the *geometry* doesn't separate your domain's concepts well.

Decision order:

1. **Start with a strong general model** (bge-large, e5, OpenAI v3). Measure recall on a golden set first.
2. **Try a domain pretrained model** if one exists (BioBERT/PubMedBERT for biomed, CodeBERT/jina-code for code).
3. **Fine-tune with contrastive pairs from your data** — mined (query, clicked-doc) pairs, or synthetic (LLM generates a question for each chunk → (question, chunk) positive). A few thousand pairs can lift recall@10 by 5–15 points.

```

1 from sentence_transformers import SentenceTransformer, losses, InputExample
2 from torch.utils.data import DataLoader
3
4 model = SentenceTransformer("BAAI/bge-base-en-v1.5")
5 train = [InputExample(texts=[q, positive_chunk]) for q, positive_chunk in mined_pairs]
6 loader = DataLoader(train, shuffle=True, batch_size=64)
7 loss = losses.MultipleNegativesRankingLoss(model) # in-batch negatives
8 model.fit(train_objectives=[(loader, loss)], epochs=1, warmup_steps=100)

```

The hidden cost: fine-tuning changes the space. New vectors are *not comparable* to old ones, so you must **re-embed your entire corpus** with the new model. At 100M chunks and 1000 chunks/sec/GPU that's ~28 GPU-hours per pass — a real operational event.

5.6 Multimodal --- CLIP and shared spaces

CLIP (Contrastive Language–Image Pretraining) trains an image encoder and a text encoder *jointly* with a contrastive loss over (image, caption) pairs, so an image and its caption land at the **same point** in a shared 512/768-dim space. This enables cross-modal search: embed a text query and retrieve images, or embed an image and retrieve captions, all with one ANN index.

```

1 # Search images with a text query — one shared space
2 image_vecs = clip.encode_image(images) # stored offline
3 q = clip.encode_text("a golden retriever on a beach")
4 hits = index.search(q, k=10) # returns nearest IMAGE vectors

```

The same recipe gives audio–text (CLAP), video–text, and document–layout embeddings. **Interview point:** multimodal RAG over PDFs with figures uses this — embed page images and surrounding text into one space.

5.7 Embedding drift & the cost of re-embedding

Two distinct problems share the name "drift":

1. **Model drift (you change models):** A new model version → entirely new geometry → **full corpus re-embedding required** + index rebuild. Plan blue/green: build the new index alongside the old, validate recall on the golden set, then cut over. Never mix vectors from two models in one index.
2. **Data/semantic drift (the world changes):** New slang, new products, new entities appear that the embedding model (trained at an earlier cutoff) represents poorly. Symptoms: recall degrades over time on fresh queries. Mitigations: periodic fine-tuning, hybrid search (BM25 catches new exact terms the model never saw), and monitoring recall on a rolling golden set.

```

1 Re-embed cost model (100M chunks):
2   throughput = 1,000 chunks/sec/GPU (bge-base on an A10)
3   total time = 100,000,000 / 1,000 = 100,000 GPU-sec ≈ 27.8 GPU-hours
4   with 8 GPUs ≈ 3.5 wall-clock hours + index build
5   API cost (if using OpenAI v3-small @ ~$0.02/1M tokens, ~300 tok/chunk)
6     ≈ 100M × 300 / 1M × $0.02 = $600 per full re-embed

```

Interview takeaway: Re-embedding is the single biggest hidden operational cost of a vec-

tor system. Always mention it when someone proposes "just upgrade the embedding model."

6 Similarity & Distance Metrics

The index needs a notion of "closeness." Three matter; one dominates text.

6.1 Cosine similarity

Measures the **angle** between two vectors, ignoring magnitude.

```
1 cos(a, b) = (a · b) / (||a|| · ||b||)
2           = Σ a_i b_i / (sqrt(Σ a_i^2) · sqrt(Σ b_i^2))
3
4 Range: [-1, 1] (1 = identical direction, 0 = orthogonal, -1 = opposite)
5 Cosine distance = 1 - cos(a, b)
```

Why cosine is the text default: Document length / token count inflates raw vector magnitude, but meaning is in the *direction*. Two paraphrases of different lengths should match — cosine ignores the length difference and compares semantic direction. Almost every sentence-transformer is trained with cosine objectives, so the geometry is calibrated for it.

6.2 Dot product (inner product)

```
1 a · b = Σ a_i b_i = ||a|| ||b|| cos(a, b)
```

Cosine **scaled by both magnitudes**. Use it when magnitude is meaningful (e.g., recommendation models that bake "popularity" or "confidence" into the norm). MIPS = Maximum Inner Product Search.

6.3 The normalized → cosine = dot equivalence (know this cold)

If both vectors are unit length ($\|a\| = \|b\| = 1$):

$$a \cdot b = \|a\| \|b\| \cos(a, b) = 1 \cdot 1 \cdot \cos(a, b) = \cos(a, b)$$

So: **normalize once, then use the cheap dot-product metric, and you get exact cosine ranking**. This is why production systems L2-normalize at embed time and configure the index for inner product. It saves the per-query norm computation.

6.4 Euclidean / L2 distance

$$L2(a, b) = \sqrt{\sum (a_i - b_i)^2}$$

Straight-line distance. For **unit-normalized** vectors, L2 and cosine produce the *same ranking* (they're monotonically related):

$$\|a - b\|^2 = \|a\|^2 + \|b\|^2 - 2(a \cdot b) = 1 + 1 - 2 \cos(a, b) = 2 - 2 \cos(a, b)$$

→ smaller L2 ⇔ larger cosine. Same nearest neighbors.

So on normalized text vectors, cosine / dot / L2 all rank identically — pick whichever your

DB optimizes. On *un-normalized* vectors they differ, and L2 is sensitive to magnitude.

6.5 Manhattan (L1)

$$L1(a, b) = \sum |a_i - b_i|$$

Sum of absolute differences. Rare in semantic search (less aligned with how embeddings are trained), more common in some recommender and tabular settings. Mention it for completeness; don't default to it for text.

6.6 Which metric, when

Metric	Use when	Notes
Cosine	Text embeddings (default)	Direction = meaning; length-invariant
Dot / IP	Normalized vectors (= cosine, cheaper) OR magnitude encodes signal (recsys)	MIPS; OpenAI/Cohere vectors normalized
L2 (Euclidean)	Image embeddings, normalized text (same ranking as cosine)	FAISS default metric
L1 (Manhattan)	Some tabular / recsys cases	Uncommon for text

Interview takeaway: For text, say "cosine — but since I normalize, I configure the index for inner product, which gives identical ranking with less compute." That one sentence signals you understand the equivalence.

7 Approximate Nearest Neighbor (ANN) Indexes

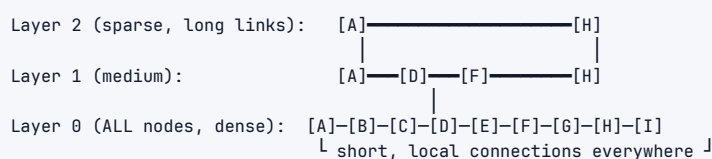
The heart of the system. Each index makes a different bet about how to avoid scanning all N vectors. Know HNSW deeply; know IVF and PQ well enough to combine them.

7.1 HNSW --- Hierarchical Navigable Small World (the default)

Intuition: skip lists + small-world graphs. A skip list speeds up linked-list search by adding sparse "express lanes" on top. HNSW does the same for nearest-neighbor search: it builds a multi-layer graph where the top layers have few nodes with long-range links (express lanes) and the bottom layer has every node with short-range links (local roads). You enter at the top, take big hops to get near the answer, then drop down to refine.

A **navigable small-world (NSW)** graph is one where greedy "always move to the neighbor closest to the query" reaches the true nearest neighbor in a few hops — like the "six degrees of separation" property of social networks. HNSW makes it *hierarchical* for $O(\log N)$ navigation.

HNSW layered structure (query q enters at the top):



Greedy search for q:

1. Enter at A (top layer). Move to whichever neighbor is closest to q.

- ```

A → H (H is closer to q at layer 2). No closer neighbor → descend.
2. Layer 1 from H: check H's neighbors (F, ...). F closer → move to F.
 No closer → descend.
3. Layer 0 from F: explore E, G, ... maintain a candidate set of size ef_search,
 greedily expand the closest unvisited, stop when no improvement.
4. Return the top-k from the candidate set.

```

Total hops  $\approx O(\log N)$ . Each hop touches a handful of neighbors.

**Greedy search walkthrough (concrete):** Suppose  $q$  is closest to node  $E$ . At layer 2 we hop  $A \rightarrow H$  (overshoot, but cheap). At layer 1 we hop  $H \rightarrow F$  (getting warmer). At layer 0 we explore  $F$ 's neighborhood, keep the `ef_search` best candidates in a priority queue, expand the closest, discover  $E$ , find nothing closer, and return  $E$  (plus the next-best as the rest of top- $k$ ). We touched maybe 50–200 nodes out of 100M.

**Construction.** Insert nodes one at a time. Each new node picks a random maximum layer (geometric distribution — most nodes only exist at layer 0, few reach the top). For each layer from its top down to 0, it runs the same greedy search to find its  $M$  nearest existing nodes and links to them. `ef_construction` controls how thorough that search is during building.

#### Parameters:

| Param               | Meaning                                 | Higher value →                           | Typical                         |
|---------------------|-----------------------------------------|------------------------------------------|---------------------------------|
| <b>M</b>            | Max neighbors per node (graph degree)   | Better recall, more memory, slower build | 16–48                           |
| <b>ef_construct</b> | Candidate list size during <i>build</i> | Better graph quality, slower build       | 100–400                         |
| <b>ef_search</b>    | Candidate list size during <i>query</i> | Higher recall, higher latency            | 50–500 (tunable at query time!) |

The killer feature: **ef\_search is a query-time knob**. You build the graph once, then dial recall vs. latency per query without rebuilding — set `ef_search=64` for a fast path and `ef_search=512` for a high-recall path.

**Memory cost.** HNSW stores the full vectors *plus* the graph edges:

```

Per vector: d × 4 bytes (float32) + M × ~8 bytes (neighbor links, both directions ~2M)
For d=768, M=32:
 vector = 768 × 4 = 3,072 bytes
 graph ≈ 2 × 32 × 8 = 512 bytes
 total ≈ 3,584 bytes/vector
100M vectors ≈ 358 GB RAM. ← This is why HNSW alone doesn't scale to billions cheaply.

```

**Why it's the default:** highest recall at a given latency, no training step, supports incremental inserts, query-time recall knob. Used by Pinecone, Weaviate, Qdrant, Milvus, pgvector, Elasticsearch/OpenSearch, Lucene. **Cons:** memory-hungry (stores full vectors), deletes are awkward (tombstones; graph degrades, eventually rebuild), build is slower than IVF.

## 7.2 IVF --- Inverted File (cluster-and-probe)

**Intuition:** Partition the space into clusters once; at query time, only search the few clusters near the query instead of all  $N$  vectors.

#### Mechanism (Voronoi cells via k-means):

1. **Train** a coarse quantizer: run k-means on a sample to find  $n_{list}$  centroids (e.g., 4096). Each centroid "owns" a Voronoi cell.
2. **Assign** every vector to its nearest centroid → an inverted list per centroid (centroid → [vector ids in that cell]).

3. **Query:** find the `nprobe` centroids nearest the query, then brute-force search only the vectors in those `nprobe` lists.

Space partitioned into `nlist=8` Voronoi cells (centroids ●):



`q` = query  
`nprobe=2` → search cells C5 and C6 only  
 (the two centroids nearest `q`),  
 skip the other 6 cells entirely.

### Parameters:

| Param         | Meaning                        | Trade-off                                                                                                                                                     |
|---------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>nlist</b>  | Number of clusters (centroids) | More cells = fewer vectors/cell = faster probe, but need higher <code>nprobe</code> for recall. Rule of thumb: $nlist \approx \sqrt{N}$ to $4 \cdot \sqrt{N}$ |
| <b>nprobe</b> | Cells searched per query       | Higher = better recall, slower. Query-time knob, like <code>ef_search</code>                                                                                  |

`N` = 100M, `nlist` = 16,384 → ~6,100 vectors per cell on average  
`nprobe` = 32 → search  $\sim 32 \times 6,100 = \sim 195,000$  vectors (vs 100M brute force)  
 → ~500× fewer distance computations

**Training requirement (the catch):** IVF must be *trained* (k-means) on a representative sample before you can add vectors. If your data distribution shifts, the cells become unbalanced (some huge, some empty) and recall drops — you must retrain. HNSW needs no training. The **edge-of-cell problem**: a query near a cell boundary may have its true nearest neighbor in an adjacent, un-probed cell — that's a recall miss, mitigated by higher `nprobe`.

**When to use:** very large datasets where HNSW memory is too high, and you can tolerate a training step. Almost always combined with **PQ** (next) to cut memory.

## 7.3 Product Quantization (PQ) --- compressing vectors

**The problem PQ solves:** storing full float32 vectors costs too much RAM at billion scale (a  $1B \times 768 \times 4B$  index = 3 TB). PQ compresses each vector to a few bytes.

### Mechanism:

1. **Split** each `d`-dim vector into `m` sub-vectors (e.g., `d=768`, `m=96` → each sub-vector is 8 dims).
2. For each of the `m` sub-spaces, run k-means to learn a **codebook** of `k=256` centroids (so each centroid id fits in 1 byte).
3. **Encode** a vector by replacing each sub-vector with the *id* of its nearest codebook centroid → the whole vector becomes `m` bytes.

Original vector (768 float32 = 3,072 bytes):  
`[-sub1-][-sub2-]...[-sub96-]` each sub = 8 floats

After PQ (`m=96` sub-vectors, 256 centroids each):  
`[id1][id2]...[id96]` each id = 1 byte → 96 bytes total

Compression: 3,072 → 96 bytes = 32× smaller.

**Asymmetric Distance Computation (ADC):** at query time you do *not* quantize the query. Instead, you precompute a lookup table: for each sub-space, the distance from the query's sub-vector to all 256 centroids (`m` × 256 distances). Then the approximate distance to any stored code is just `m` table lookups + adds — no full distance math. This keeps accuracy higher than quantizing both sides ("symmetric").

```

1 PQ distance(query, code):
2 precompute LUT[m][256] = dist(query_sub_j, centroid_c) for all j, c # once per query
3 for each stored vector with codes [c1..cm]:
4 approx_dist = LUT[0][c1] + LUT[1][c2] + ... + LUT[m-1][cm] # m adds

```

### The memory math (the number to quote):

```

1 BILLION vectors, d=768:
float32 (no compression): 1e9 × 768 × 4 B = 3.07 TB (won't fit in RAM)
PQ, m=96, 8-bit codes: 1e9 × 96 × 1 B = 96 GB (fits on one big box)
→ 32× memory reduction.
Cost: recall drops (lossy compression). Typically recall@10 ~0.85-0.95
 depending on m, recoverable by re-ranking the PQ candidates with exact
 distances on the float vectors (if kept on disk/SSD).

```

**IVF-PQ (the workhorse for billion-scale):** combine them — IVF narrows to nprobe cells (avoid scanning all N), PQ compresses the vectors in those cells (fit in RAM). This is the standard FAISS recipe for billion-vector indexes (IndexIVFPQ). A refinement, **IVF-PQ + reranking** (IndexIVFPQR / 0PQ): retrieve top-100 by PQ approx distance, then re-score those 100 with exact float distances for final ranking.

```

1 import faiss
2 d, nlist, m, nbits = 768, 16384, 96, 8
3 quantizer = faiss.IndexFlatIP(d) # coarse quantizer (centroids)
4 index = faiss.IndexIVFPQ(quantizer, d, nlist, m, nbits)
5 index.train(sample_vectors) # k-means for IVF + PQ codebooks
6 index.add(all_vectors) # stored as PQ codes
7 index.nprobe = 32
8 D, I = index.search(query, k=10)

```

## 7.4 LSH --- Locality-Sensitive Hashing

**Intuition:** design hash functions so that *similar* vectors collide (land in the same bucket) with high probability. Query by hashing the query and only scanning its bucket(s).

For cosine, use **random hyperplane hashing**: pick  $r$  random hyperplanes; each vector's hash bit is the sign of its dot product with that hyperplane's normal. Vectors on the same side of all hyperplanes share a hash → likely similar.

```

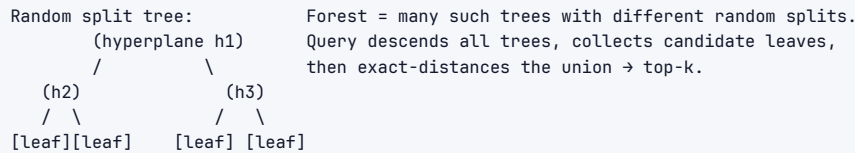
hash(v) = (sign(v·r1), sign(v·r2), ..., sign(v·rb)) # b bits, b random hyperplanes
Similar vectors → same signs → same bucket → small candidate set to brute-force.
Multiple hash tables (L of them) raise recall (a miss in one table may hit another).

```

**Trade-off:** simple, theoretically clean, supports streaming inserts. But in practice it needs many tables/bits to hit high recall, which costs memory, and it generally loses to HNSW on the recall-per-byte and recall-per-millisecond curves. Mostly of historical/theoretical importance now; still used for near-duplicate detection (MinHash for sets).

## 7.5 Tree-based --- Annoy

**Annoy** (Spotify) builds a **forest of random projection trees**. Each tree recursively splits the dataset by a random hyperplane; a leaf holds a small bucket. Querying descends multiple trees and unions the leaf candidates.



**Trade-off:** memory-mapped files → great for **read-only**, shareable indexes (Spotify serves the same index across many machines via mmap). But the index is **immutable** — you rebuild to add data. Lower recall-per-memory than HNSW. Good when you have a static corpus and want simple, mmap-able files.

## 7.6 ScaNN --- Anisotropic quantization (Google)

**ScaNN's insight:** standard PQ minimizes *reconstruction* error (how well the code approximates the original vector). But for **maximum inner product search** what actually matters is preserving the *inner product with likely query directions*. ScaNN uses **anisotropic vector quantization** — it weights quantization error by how much it distorts inner products in the directions that matter, penalizing errors parallel to the vector more than orthogonal ones. Result: higher recall at the same compression. Combined with a fast SIMD-optimized scan, it tops the ann-benchmarks recall/latency frontier for in-memory MIPS.

**Trade-off:** state-of-the-art speed/recall, but more complex to tune; strongest as a library (or via Vertex AI) rather than a turnkey DB.

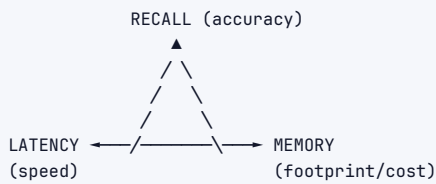
## 7.7 ANN comparison table

| Index             | Build time | Query speed      | Recall                       | Memory                     | Inserts                       | Training?                 | Best for                         |
|-------------------|------------|------------------|------------------------------|----------------------------|-------------------------------|---------------------------|----------------------------------|
| <b>HNSW</b>       | Slow       | Very fast        | Very high                    | High (full vecs + graph)   | Incremental                   | No                        | Default; <~100M, recall-critical |
| <b>IVF (Flat)</b> | Fast       | Fast             | High (tune nprobe)           | Full vecs                  | Add anytime; retrain on drift | Yes (k-means)             | Large, memory-ok                 |
| <b>IVF-PQ</b>     | Moderate   | Fast             | Moderate-high                | <b>Very low</b> (PQ codes) | Add anytime                   | Yes (k-means + codebooks) | Billion-scale, RAM-bound         |
| <b>PQ (flat)</b>  | Moderate   | Fast             | Moderate                     | Very low                   | Add anytime                   | Yes (codebooks)           | Memory-extreme                   |
| <b>LSH</b>        | Fast       | Fast             | Moderate (needs many tables) | Moderate-high              | Streaming                     | No                        | Near-dup detection, theory       |
| <b>Annoy</b>      | Moderate   | Fast             | Moderate                     | Low (mmap)                 | <b>Immutable</b> (rebuild)    | No                        | Static, read-only, mmap-shared   |
| <b>ScaNN</b>      | Moderate   | <b>Very fast</b> | Very high                    | Low-moderate               | Rebuild-ish                   | Yes                       | In-memory MIPS at scale (Google) |

**Interview takeaway:** Default to **HNSW** for most workloads (<100M, recall matters, RAM available). Reach for **IVF-PQ** when memory is the binding constraint (billions of vectors). Mention **ScaNN** if asked about Google or the absolute speed/recall frontier. Mention **Annoy** for static, mmap-shared, read-only indexes.

## 8 The Recall / Latency / Memory Trade-off

Every ANN system lives inside a triangle. You cannot maximize all three; you pick a point.



HNSW with high M, high ef\_search → high recall, high memory, moderate latency  
 IVF-PQ with low m, low nprobe → low memory, low latency, lower recall  
 Brute force → perfect recall, huge latency, full memory

### 8.1 Recall@k --- the metric

Recall@k = of the *true* top-k nearest neighbors (per exact brute force), what fraction did the ANN index return in *its* top-k?

True top-10 for a query: {d3, d7, d9, d11, d14, d20, d22, d31, d40, d55}  
 ANN returned top-10: {d3, d7, d9, d11, d14, d20, d22, d31, d40, d99}  
 9 of 10 correct → recall@10 = 0.90

You compute it by running exact kNN on a sample of queries (the ground truth) and comparing. Target recall depends on the product: web search tolerates ~0.90; medical/legal retrieval may demand ~0.99.

### 8.2 How parameters tune the triangle

| Want      | HNSW lever                                  | IVF / IVF-PQ lever                                   | Side effect               |
|-----------|---------------------------------------------|------------------------------------------------------|---------------------------|
| ↑ Recall  | ↑ ef_search, ↑ M                            | ↑ nprobe, ↑ nlist resolution, ↓ PQ compression (↑ m) | ↑ latency and/or ↑ memory |
| ↓ Latency | ↓ ef_search                                 | ↓ nprobe                                             | ↓ recall                  |
| ↓ Memory  | (limited — vecs are full)<br>use PQ variant | use PQ (↓ m bytes), 8-bit→4-bit                      | ↓ recall                  |

#### Concrete tuning scenario (100M, d=768, HNSW):

M=16, ef\_search=64 → recall@10 ≈ 0.92, p95 latency ≈ 8 ms, ~330 GB RAM  
 M=32, ef\_search=128 → recall@10 ≈ 0.97, p95 latency ≈ 15 ms, ~358 GB RAM  
 M=48, ef\_search=512 → recall@10 ≈ 0.99, p95 latency ≈ 40 ms, ~390 GB RAM

#### Concrete tuning scenario (1B, d=768, IVF-PQ):

m=96 (8-bit), nprobe=16 → recall@10 ≈ 0.82, p95 ≈ 6 ms, ~96 GB RAM  
 m=96, nprobe=64 → recall@10 ≈ 0.90, p95 ≈ 18 ms, ~96 GB RAM  
 m=96, nprobe=64 + rerank top-200 with float vecs on SSD → recall@10 ≈ 0.96, p95 ≈ 30 ms

**The reranking trick to break the triangle:** retrieve a *larger* candidate set cheaply (high speed, lower per-candidate accuracy), then re-score the small candidate set with exact distances (or a cross-encoder). You buy back recall/precision without paying full-index cost.

**Interview takeaway:** Never say "I'll use HNSW" and stop. Say: "HNSW with M=32, and I'd tune ef\_search against a recall@10 target of 0.97 measured on a golden set, accepting ~15 ms

p95 and ~358 GB RAM — if memory is the constraint I'd switch to IVF-PQ and rerank.” That's the senior answer.

## 9 Building a RAG / Semantic-Search Pipeline

RAG = Retrieval-Augmented Generation: retrieve relevant chunks from a vector store, inject them into an LLM prompt, generate a grounded answer. Semantic search is the same pipeline minus the generation step. Retrieval quality dominates final quality — “garbage retrieved, garbage generated.”

```
INGEST (offline): Docs → Parse → Chunk → Embed → Upsert(vector + metadata)
QUERY (online): Query → Embed → ANN retrieve top-k (+ BM25) → Fuse → Rerank → top 3-5 → LLM
```

### 9.1 Chunking strategies

You don't embed whole documents — you embed **chunks**, because (a) embedding models have token limits (256–512 tokens is the trained sweet spot for many), and (b) retrieval precision needs the *relevant passage*, not a 50-page PDF.

| Strategy                          | How                                                                                                 | Best for               | Pitfall                         |
|-----------------------------------|-----------------------------------------------------------------------------------------------------|------------------------|---------------------------------|
| <b>Fixed-size</b>                 | Every N tokens (e.g., 512)                                                                          | Quick start            | Cuts mid-sentence, splits ideas |
| <b>Recursive character</b>        | Split on <code>\n\n</code> , then <code>\n</code> , then <code>.</code> , then space until $\leq N$ | General text (default) | Still misses semantic units     |
| <b>Sentence / paragraph</b>       | Split on sentence/para boundaries                                                                   | FAQs, articles         | Uneven sizes                    |
| <b>Semantic chunking</b>          | Embed sentences; split where consecutive similarity drops                                           | Dense technical docs   | Expensive (embed to chunk)      |
| <b>Structural (Markdown/HTML)</b> | Split on headers H1/H2/H3, code blocks, tables                                                      | Docs, wikis            | Uneven chunk sizes              |
| <b>Parent-child</b>               | Retrieve small child chunk, feed larger parent to LLM                                               | Precision + context    | Two-level indexing              |

**Overlap.** Adjacent chunks share a sliding window (e.g., 50–100 tokens) so a sentence split across a boundary still appears whole in at least one chunk. Costs storage (more chunks) but prevents “answer cut in half” misses.

#### The size trade-off (with examples):

```
Chunk too SMALL (e.g., 64 tokens):
 Chunk = "Click Settings, then Security."
 Query = "how do I enable two-factor authentication?"
 → retrieved, but lacks the surrounding steps → LLM can't answer fully. (low context)

Chunk too LARGE (e.g., 2000 tokens):
 Chunk = entire 8-page "Account Security" article.
 Query = "reset password" → chunk matches, but 95% is irrelevant text that
 dilutes the embedding (averaged meaning) AND wastes the LLM context window.
 The relevant 2 sentences are buried. (low precision, high cost)

Sweet spot: 256-512 tokens, 10-20% overlap.
```

```

1 from langchain_text_splitters import RecursiveCharacterTextSplitter
2
3 splitter = RecursiveCharacterTextSplitter(
4 chunk_size=512, chunk_overlap=64, # ~512 tokens, ~12% overlap
5 separators=["\n\n", "\n", ". ", " ", ""], # prefer semantic boundaries
6)
7 chunks = splitter.split_text(document_text)

```

**Metadata is non-negotiable** — store it alongside every vector for filtering and citation:

```

1 record = {
2 "id": "doc42#chunk7",
3 "vector": embed(chunk_text),
4 "metadata": {
5 "text": chunk_text,
6 "source": "security_guide_v3.pdf",
7 "page": 12, "section": "Two-Factor Auth",
8 "tenant_id": "acme", "acl": ["support", "admin"],
9 "created_at": "2025-02-01", "doc_id": "doc42",
10 },
11 }

```

## 9.2 Embed + upsert

Use the **same model** for ingest and query (different models → incompatible geometry → garbage retrieval). Normalize. Upsert in batches.

```

1 import itertools
2 def batched(it, n):
3 it = iter(it)
4 while batch := list(itertools.islice(it, n)):
5 yield batch
6
7 for batch in batched(chunks, 256):
8 vecs = model.encode([c.text for c in batch], normalize_embeddings=True)
9 index.upsert([(c.id, v, c.metadata) for c, v in zip(batch, vecs)])

```

## 9.3 Retrieval top-k

Retrieve more than you'll show the LLM (e.g., top-20–50), then narrow via fusion/rerank. Over-retrieving improves the chance the right chunk is in the candidate set.

```

1 q = model.encode(user_query, normalize_embeddings=True)
2 candidates = index.search(q, k=30, filter={"tenant_id": "acme"})

```

## 9.4 Metadata filtering --- pre- vs post-filter (a real correctness/perf trap)

You often must restrict results ("only this tenant," "only docs the user can see," "only after 2024"). There are two ways, and the difference is a classic interview gotcha:

**POST-FILTER:** ANN returns top-k by vector distance, THEN drop those failing the predicate.  
 Problem: if only 1% of vectors match the filter, your top-30 might contain ZERO matches  
 → you return nothing, even though good matches exist deeper in the ranking.  
 Correctness bug. To compensate you over-fetch (k=3000), which is slow.

**PRE-FILTER:** Restrict the candidate set to matching vectors FIRST, then do ANN within it.  
 Correct results, but naive pre-filtering breaks the ANN graph (HNSW navigates through nodes that may be filtered out → can't traverse → recall collapses).

**PRODUCTION ANSWER:** "filtered ANN" – the index integrates the predicate INTO traversal.  
 Qdrant: payload index + filterable HNSW. Weaviate/Milvus: similar. pgvector: combine with a B-tree/partial index, or partition by tenant. For very selective filters (a handful of matches), fall back to exact brute force over the matching subset – it's tiny.

```

1 # Qdrant filtered search – predicate is applied DURING graph traversal
2 from qdrant_client.models import Filter, FieldCondition, MatchValue
3 hits = client.search(
4 collection_name="docs",
5 query_vector=q,
6 query_filter=Filter(must=[FieldCondition(key="tenant_id", match=MatchValue(value="acme"))]),
7 limit=30,
8)

```

**Interview takeaway:** When asked "how do you restrict search to a tenant?" the wrong answer is "filter the results afterward." The right answer names the post-filter recall bug and proposes filtered-ANN (predicate fused into traversal) or partitioning, with exact brute force for highly selective filters.

## 9.5 Hybrid search --- dense + sparse (BM25) with RRF

Dense embeddings miss exact strings (SKUs, error codes, rare names); BM25 nails them. BM25 misses paraphrase; dense nails it. Run **both** and fuse.

**Reciprocal Rank Fusion (RRF)** combines ranked lists using only the *rank* (not the raw, incomparable scores):

```
RRF_score(doc) = Σover_lists 1 / (k + rank_in_that_list) # k ≈ 60 by convention
```

A doc ranked #1 by BM25 and #3 by dense:  $1/(60+1) + 1/(60+3) = 0.0164 + 0.0159 = 0.0323$   
 A doc ranked #2 by dense only:  $1/(60+2) = 0.0161$   
 → The doc both methods like ranks highest. Robust, no score normalization needed.

```

1 def rrf(rank_lists, k=60, top_n=20):
2 scores = {}
3 for ranked in rank_lists: # each is [doc_id, ...] best-first
4 for rank, doc_id in enumerate(ranked):
5 scores[doc_id] = scores.get(doc_id, 0.0) + 1.0 / (k + rank)
6 return sorted(scores, key=scores.get, reverse=True)[:top_n]
7
8 dense_hits = [h.id for h in index.search(q, k=50)]
9 bm25_hits = [h.id for h in bm25.search(user_query, k=50)]
10 fused = rrf([dense_hits, bm25_hits])

```

## 9.6 Reranking --- the two-stage retrieve → rerank pattern

Retrieval uses a **bi-encoder** (query and doc embedded *independently*) – fast, scalable, but it never lets the query and document "see" each other, so it's coarse. A **cross-encoder** feeds

[query, doc] *together* through a transformer with full cross-attention and outputs a single relevance score — far more accurate, but  $O(\text{candidates})$  forward passes, so you can't run it over the whole corpus.

**The two-stage architecture resolves this:** retrieve top-30–50 cheaply with the bi-encoder ANN index, then **rerank just those 30–50** with the cross-encoder and keep the top 3–5 for the LLM.

Stage 1 (recall): bi-encoder ANN over 100M docs → top 50 candidates (fast, coarse)  
 Stage 2 (precision): cross-encoder scores 50 (query,doc) pairs → top 5 (slow, accurate)  
 50 forward passes is cheap; 100M would be impossible.

```

1 from sentence_transformers import CrossEncoder
2 reranker = CrossEncoder("cross-encoder/ms-marco-MiniLM-L-6-v2")
3
4 def retrieve_and_rerank(query, top_n=5):
5 q = model.encode(query, normalize_embeddings=True)
6 candidates = index.search(q, k=50) # bi-encoder, fast
7 pairs = [(query, c.metadata["text"]) for c in candidates]
8 scores = reranker.predict(pairs) # cross-encoder, accurate
9 ranked = [c for _, c in sorted(zip(scores, candidates), key=lambda x: -x[0])]
10 return ranked[:top_n]

```

Commercial rerankers: **Cohere Rerank**, **Voyage rerank**, **Jina Reranker**. Reranking 50 docs adds ~30–80 ms but lifts answer quality substantially — almost always worth it in production.

## 9.7 Putting it together --- query pipeline

```

1 def rag_answer(user_query, tenant_id, user_acl):
2 q = model.encode(user_query, normalize_embeddings=True)
3 dense = index.search(q, k=50, filter={"tenant_id": tenant_id, "acl": {"$in": user_acl}})
4 bm25 = bm25_index.search(user_query, k=50, filter={"tenant_id": tenant_id})
5 fused_ids = rrf([h.id for h in dense], [h.id for h in bm25], top_n=50)
6 fused = [lookup(i) for i in fused_ids]
7 top = rerank(user_query, fused, top_n=5)
8 context = "\n\n".join(
9 f"[{c.metadata['source']}] p[{c.metadata['page']}] \n{c.metadata['text']}" for c in top
10)
11 return llm.generate(
12 f"Answer using ONLY this context. Cite sources.\n\n{context}\n\nQ: {user_query}"
13)

```

## 10 Vector Database Systems

You can use a **library** (FAISS, ScaNN, Annoy — you manage storage, sharding, persistence, filtering yourself) or a **database** (Pinecone, Weaviate, Milvus, Qdrant, pgvector — they add meta-data, filtering, replication, APIs, persistence).

## 10.1 pgvector (Postgres extension) --- the "use what you have" choice

If you're already on Postgres, pgvector adds a vector column type plus HNSW and IVF indexes. You get SQL joins, transactions, existing backups/replication, and ACID filtering *for free*.

```

1 CREATE EXTENSION IF NOT EXISTS vector;
2
3 CREATE TABLE chunks (
4 id bigserial PRIMARY KEY,
5 doc_id text,
6 tenant_id text,
7 page int,
8 text text,
9 embedding vector(768) -- 768-dim dense vector
10);
11
12 -- HNSW index (cosine). vector_ip_ops for inner product, vector_l2_ops for L2.
13 CREATE INDEX ON chunks
14 USING hnsw (embedding vector_cosine_ops)
15 WITH (m = 16, ef_construction = 64);
16
17 -- Partial index for a tenant (helps filtered search)
18 CREATE INDEX ON chunks (tenant_id);
19
20 -- Query: top-10 nearest, filtered by tenant (pre-filter via WHERE + index)
21 SET hnsw.ef_search = 100; -- query-time recall knob
22 SELECT id, doc_id, page, text,
23 1 - (embedding <=> $1) AS cosine_similarity -- <=> is cosine distance
24 FROM chunks
25 WHERE tenant_id = 'acme'
26 ORDER BY embedding <=> $1 -- <=> cosine, <#> inner product, <-> L2
27 LIMIT 10;

```

$\Leftrightarrow$  = cosine distance,  $\langle \# \rangle$  = negative inner product,  $\Leftrightarrow$  = L2. **Caveat:** at very high N (>50–100M) and high-selectivity filters, pgvector's HNSW can lose recall under tight WHERE filters; partition by tenant or move to a dedicated vector DB.

## 10.2 The major systems

| DB              | Type         | Index                      | Scale           | Filtering                                  | Highlights                          | Best for                     |
|-----------------|--------------|----------------------------|-----------------|--------------------------------------------|-------------------------------------|------------------------------|
| <b>pgvector</b> | Postgres ext | HNSW, IVFFlat              | ~10s of M       | SQL WHERE (pre-filter)                     | SQL joins, ACID, reuse infra        | Already on Postgres          |
| <b>Pinecone</b> | Managed SaaS | proprietary (HNSW-like)    | Billion         | Metadata filters                           | Zero ops, auto-scale, serverless    | Fast to production           |
| <b>Weaviate</b> | OSS + Cloud  | HNSW                       | Millions B      | Strong, filterable HNSW                    | GraphQL, modules, hybrid built-in   | Feature-rich OSS, hybrid     |
| <b>Qdrant</b>   | OSS + Cloud  | HNSW                       | Million B       | <b>Excellent</b> payload + filterable HNSW | Rust, fast, quantization built-in   | Filtered search, performance |
| <b>Milvus</b>   | OSS + Cloud  | HNSW, IVF-PQ, DiskANN, GPU | <b>Billions</b> | Yes                                        | Most scalable OSS, many index types | Massive self-hosted          |

| DB                                | Type          | Index                        | Scale      | Filtering | Highlights                         | Best for                 |
|-----------------------------------|---------------|------------------------------|------------|-----------|------------------------------------|--------------------------|
| <b>FAISS</b>                      | Library       | Flat, HNSW, IVF, IVF-PQ, OPQ | Billion    | DIY       | Max control, GPU, all algorithms   | Research, custom infra   |
| <b>ScaNN</b>                      | Library       | Anisotropic + tree           | Billions   | DIY       | Fastest in-memory MIPS (Google)    | Speed-critical MIPS      |
| <b>Annoy</b>                      | Library       | RP-tree forest               | Millior    | DIY       | mmap, immutable, simple            | Static read-only, recsys |
| <b>Elasticsearch / OpenSearch</b> | Search engine | HNSW (Lucene)                | Millions/B | Native    | Best hybrid (BM25 + vector in one) | Already on ES, hybrid    |

### 10.3 Managed vs self-hosted

|                       | Managed (Pinecone, Weaviate Cloud, Qdrant Cloud) | Self-hosted (Milvus, Qdrant, FAISS)     |
|-----------------------|--------------------------------------------------|-----------------------------------------|
| <b>Ops burden</b>     | Near zero                                        | You run sharding, replication, upgrades |
| <b>Cost</b>           | \$ per pod/usage (can be high at scale)          | Hardware + eng time                     |
| <b>Control</b>        | Limited tuning                                   | Full (custom indexes, GPU)              |
| <b>Data residency</b> | Their cloud (compliance concerns)                | Your infra                              |
| <b>Scale ceiling</b>  | Very high (their problem)                        | Yours to engineer                       |

**Interview takeaway:** Default recommendation — “If we're on Postgres and < ~10M vectors, pgvector. For fast time-to-production, Pinecone. For self-hosted scale with filtering, Qdrant or Milvus. For maximum control / research / GPU / billion-scale custom, FAISS. If we already run Elasticsearch and need hybrid, use its native kNN.”

## 11 Scaling, Updates & Operations

### 11.1 Sharding

At billions of vectors, one machine can't hold the index. Shard the vectors across nodes; query all shards in parallel; merge the per-shard top-k.

| Strategy                               | How                                                | Pro                                       | Con                                                  |
|----------------------------------------|----------------------------------------------------|-------------------------------------------|------------------------------------------------------|
| <b>Shard by id (random/hash)</b>       | hash(id) % num_shards                              | Even load, simple                         | <b>Every query hits every shard</b> (scatter-gather) |
| <b>Shard by cluster (IVF centroid)</b> | Route query to shards owning the nearest centroids | Query touches few shards                  | Skewed cells = hot shards; rebalancing on drift      |
| <b>Shard by tenant/metadata</b>        | One shard (or index) per tenant                    | Filtered queries hit one shard; isolation | Uneven tenant sizes; many small indexes              |

**SCATTER-GATHER (hash sharding):**

Query → [shard0 top-k][shard1 top-k]...[shardN top-k] → merge → global top-k  
 Latency = max(shard latency) + merge. Tail latency dominated by the slowest shard.

**ROUTED (cluster sharding):**

Query → nearest 2 centroids → only their 2 shards → merge.  
 Fewer shards touched → less load, but a query near a boundary may miss → lower recall.

## 11.2 Replication

Each shard is replicated (e.g., 3×) for availability and read throughput. Reads load-balance across replicas; a replica failure doesn't drop data. Writes go to a primary and propagate. Same leader-follower trade-offs as any distributed store (replication lag → a just-upserted vector may not yet be searchable on all replicas → "freshness").

## 11.3 Inserts vs deletes --- the hard part

**Inserts** are easy for HNSW (incremental graph insertion) and IVF (assign to nearest cell). The cost: HNSW insertion runs a search to find neighbors ( $\approx$  query cost), so high write rates are heavier than reads.

**Deletes** are the genuinely hard problem:

- ▶ HNSW graphs don't support clean node removal — deleting a node would orphan its neighbors' links and degrade navigability.
- ▶ The standard solution is **tombstones (soft delete)**: mark the vector deleted, keep it in the graph, filter it out of results at query time.
- ▶ Tombstones accumulate → wasted memory + degraded recall (the graph routes through dead nodes) → periodic **rebuild/compaction** to physically remove them, analogous to LSM-tree compaction.

Delete v: mark v.deleted = true (tombstone; still in graph, filtered at query)  
 When tombstone ratio > ~20-30% → rebuild index from live vectors (offline or shadow)

IVF handles updates better (just drop the id from a cell's list), but accumulated churn unbalances cells → eventual retrain.

## 11.4 Filtered search at scale (recap with the scaling angle)

The pre/post-filter problem (from the RAG section) gets worse at scale. For **selective** filters (few matches), exact brute force over the matching subset beats ANN. For **non-selective** filters, filtered-HNSW or partitioning. Partitioning by the common filter key (tenant, time bucket) is often the cleanest: each partition is its own small index, queried directly — no recall loss.

## 11.5 Re-embedding when the model changes (full rebuild)

This is the operational elephant. A new embedding model = new geometry = **every vector must be recomputed and the index rebuilt**. Old and new vectors are not comparable; you cannot incrementally migrate.

**Blue/Green re-embed (zero-downtime):**

1. Stand up a NEW index (green) alongside the live one (blue).
2. Re-embed the whole corpus with the new model → populate green.
3. Validate: run the golden query set against green; require recall@10 ≥ blue.
4. Dual-write new ingests to both during the transition.
5. Cut traffic over to green; decommission blue.

Cost (100M chunks): ~28 GPU-hours embed + index build + 2× storage during overlap.

## 11.6 Freshness

How fast must a new document be searchable? Real-time chat memory needs seconds; a docs corpus can tolerate hours. Patterns: a small **hot index** (recent inserts, frequently rebuilt or HNSW with fast inserts) merged with a large **cold index** (rebuilt nightly); queries hit both and merge. Trade freshness against rebuild cost.

**Interview takeaway:** "Deletes via tombstones + periodic rebuild; inserts incremental but ~as costly as a query; model upgrades force a full blue/green re-embed — that's the biggest operational risk, budget ~28 GPU-hours per 100M chunks." Saying this signals real operational experience.

## 12 Evaluation

You cannot tune recall/latency/memory without measuring. Build a **golden set** and track retrieval metrics offline, then confirm online.

### 12.1 Building a golden set

A golden set is a list of (query, relevant\_doc\_ids) pairs — the ground truth of what *should* be retrieved.

Sources: hand-labeled by experts; mined from click logs (clicked results = relevant); or **synthetically generated** (LLM reads each chunk and writes a question it answers → (question, chunk\_id) is a positive pair). Aim for a few hundred to a few thousand queries spanning head (common) and tail (rare) cases.

### 12.2 Retrieval metrics

| Metric             | Formula / idea                                             | Captures                               |
|--------------------|------------------------------------------------------------|----------------------------------------|
| <b>Recall@k</b>    | (relevant retrieved in top-k) / (total relevant)           | Did we <i>find</i> the right docs?     |
| <b>Precision@l</b> | (relevant in top-k) / k                                    | Are top-k <i>clean</i> (not noisy)?    |
| <b>MRR</b>         | mean of 1/rank of the first relevant result                | How high is the <i>first</i> good hit? |
| <b>nDCG@k</b>      | DCG / ideal DCG, with graded relevance & position discount | Ranking quality with graded relevance  |
| <b>Hit@k</b>       | fraction of queries with $\geq 1$ relevant in top-k        | Coarse "did we get anything right?"    |

MRR example: query A first-relevant at rank 1 → 1/1  
 query B first-relevant at rank 3 → 1/3  
 query C first-relevant at rank 2 → 1/2  
 $MRR = (1 + 0.333 + 0.5) / 3 = 0.611$

nDCG@k:  $DCG = \sum_i rel_i / \log_2(i+1)$  (rel can be graded 0/1/2/3)  
 $nDCG = DCG / IDCG$  (DCG of the perfect ranking). Range [0,1].  
 Rewards putting the most-relevant doc at the top, not just somewhere in top-k.

```

1 def recall_at_k(retrieved_ids, relevant_ids, k):
2 top = set(retrieved_ids[:k])
3 return len(top & set(relevant_ids)) / len(relevant_ids)
4
5 def mrr(list_of_retrieved, list_of_relevant):
6 total = 0.0
7 for retrieved, relevant in zip(list_of_retrieved, list_of_relevant):

```

```

8 rel = set(relevant)
9 for rank, doc in enumerate(retrieved, start=1):
10 if doc in rel:
11 total += 1.0 / rank
12 break
13 return total / len(list_of_retrieved)

```

### 12.3 ANN recall vs end-to-end recall (don't conflate)

Two different "recall" numbers:

- **ANN recall@k** — does the index return the *exact* nearest vectors? (Index quality; ground truth = brute force.)
- **Retrieval recall@k** — did we retrieve the *truly relevant documents*? (System quality; ground truth = golden labels.)

You can have 0.99 ANN recall but poor retrieval recall if the *embedding model* (not the index) is bad — the index faithfully returns the wrong neighbors. Diagnose by comparing ANN-vs-brute-force (isolates index) against golden-set recall (isolates whole system).

### 12.4 Offline vs online

|               | Offline                                  | Online                                                    |
|---------------|------------------------------------------|-----------------------------------------------------------|
| <b>Signal</b> | Recall@k, nDCG, MRR vs golden set        | Click-through, dwell time, answer thumbs-up, task success |
| <b>Speed</b>  | Minutes                                  | Days (statistical significance)                           |
| <b>Use</b>    | Gate every change (model/index/chunking) | A/B test the winners; catch real-world gaps               |

**For full RAG** (with generation), add faithfulness/groundedness and answer-relevance (RA-GAS, LLM-as-judge) — but retrieval recall is the upstream lever; fix it first.

**Interview takeaway:** "I'd build a golden set (synthetic + log-mined), track recall@k and nDCG offline to gate index/model/chunking changes, separate ANN recall from retrieval recall to localize regressions, and A/B test online for click-through and answer success."

## 13 Designing Search over 100M Documents (Worked Example)

**Prompt:** "Design semantic search / RAG over 100 million documents with sub-100 ms p95 latency."

Walk it as: clarify → estimate → embedding → chunking → index → sharding → hybrid+rerank → budget.

### 13.1 1. Clarify requirements

- 100M *documents*; after chunking, expect ~5 chunks/doc → \*\*~500M vectors\*\*.
- Latency target: p95 < 100 ms end-to-end. Recall target: recall@10 ≥ 0.95.
- Filters: tenant + ACL + date. Freshness: new docs searchable within minutes. Read-heavy (1000s QPS).

## 13.2 2. Back-of-envelope sizing

Vectors: 100M docs  $\times$  5 chunks = 500,000,000 chunks  
 Model: bge-base, d=768, normalized  $\rightarrow$  use inner-product metric  
 Raw float32 storage: 500M  $\times$  768  $\times$  4 B = 1.536 TB (too big for one box's RAM)

This number forces the architecture: 1.5 TB of raw vectors  $\rightarrow$  either many HNSW machines or compress with PQ. We'll do **IVF-PQ + sharding + reranking**.

## 13.3 3. Embedding model

- **bge-base-en-v1.5 (768-dim)** — strong open-source, self-hostable (no per-call API cost at 500M scale), 256–512 token sweet spot. L2-normalize  $\rightarrow$  inner-product metric.
- Fine-tune later on mined (query, clicked-chunk) pairs if golden-set recall is weak.
- Embedding cost: 500M chunks  $\div$  1000 chunks/s/GPU = 500K GPU-sec  $\approx$  139 GPU-hours (one-time, parallelizable across GPUs).

## 13.4 4. Chunking

- **Recursive character split**, 512 tokens, 64-token overlap, prefer header/paragraph boundaries (structural for the docs that have markup).
- Store metadata: {text, doc\_id, source, page, tenant\_id, acl[], created\_at}.

## 13.5 5. Index choice + params

500M vectors  $\rightarrow$  memory is the binding constraint  $\rightarrow$  **IVF-PQ** (compress) with a **rerank** stage:

IVF: nlist =  $4 \times \sqrt{500M} \approx 4 \times 22,360 \approx 90,000$  centroids  
 $\rightarrow$  ~5,500 vectors per cell. nprobe = 64 (query-time tunable).  
 PQ: d=768, m=96 sub-vectors (8 dims each), 8-bit codes (256 centroids/sub).  
 Per-vector storage: 96 bytes (+ a few bytes overhead).  
 Memory: 500M  $\times$  ~100 B  $\approx$  50 GB of PQ codes  $\leftarrow$  now fits across a few machines.  
 Rerank: keep full float vectors on NVMe SSD; for the top-200 IVF-PQ candidates,  
 re-score with exact inner product  $\rightarrow$  recovers recall@10 to ~0.96.

Compared with HNSW-flat which would need  $\sim 500M \times 3.5 \text{ KB} \approx$  **1.75 TB RAM** (many more machines) — IVF-PQ is  $\sim 35\times$  cheaper on memory, accepting a rerank step.

## 13.6 6. Sharding & replication

Shard by hash(chunk\_id) into, say, 8 shards  $\rightarrow$  ~62.5M vectors / ~6.5 GB PQ codes each.  
 Each shard: 3 replicas (availability + read throughput).  
 Query = scatter-gather: fan out to 8 shards, each returns top-50, merge  $\rightarrow$  top-200,  
 then rerank. Latency  $\approx$  max(shard) + merge + rerank.  
 For tenant isolation, optionally co-locate a tenant's chunks (route filtered queries to fewer shards). Large tenants get dedicated indexes.

## 13.7 7. Hybrid + rerank pipeline

```

Query
├─ embed (bge-base) $\rightarrow$ IVF-PQ ANN (nprobe=64) per shard $\rightarrow$ merge top-200 [~25 ms]
├─ BM25 (Elasticsearch/Lucene) $\rightarrow$ top-100 [~15 ms, parallel]
└─
RRF fuse (k=60) $\rightarrow$ top-100
└─
Cross-encoder rerank (ms-marco MiniLM) on 100 $\rightarrow$ top-5 [~40 ms]
└─
(RAG) inject top-5 into LLM prompt with citations

```

## 13.8 8. Latency & resource budget

| Python |                                           |                                    |
|--------|-------------------------------------------|------------------------------------|
| 1      | Embed query (1 forward pass, GPU/CPU)     | ~5 ms                              |
| 2      | IVF-PQ scatter-gather (8 shards parallel) | ~25 ms (p95; tail = slowest shard) |
| 3      | BM25 (parallel, overlaps)                 | ~15 ms                             |
| 4      | RRF fuse                                  | ~2 ms                              |
| 5      | SSD rerank of top-200 (exact float)       | ~10 ms                             |
| 6      | Cross-encoder rerank top-100              | ~40 ms                             |
| 7      |                                           |                                    |
| 8      | End-to-end p95                            | ~80 ms ✓ under 100 ms              |
| 9      | Recall@10 (post-rerank)                   | ~0.96 ✓                            |
| 10     | Memory (PQ codes, replicated 3x)          | ~150 GB total (50 GB × 3)          |
| 11     | One-time embed                            | ~139 GPU-hours                     |

## 13.9 9. Trade-offs to state aloud

- › **IVF-PQ vs HNSW:** chose PQ for ~35× memory savings; pay with a rerank stage and a k-means training step (retrain on drift). If memory weren't constrained, HNSW (M=32) would give ~0.97 recall with no rerank but ~1.75 TB RAM.
- › **nprobe / ef\_search** are the live recall-vs-latency knobs.
- › **Filtering:** filtered-IVF or tenant partitioning; exact brute force for highly selective ACL filters.
- › **Model upgrade:** blue/green full re-embed (~139 GPU-hours), validate on golden set, cut over.
- › **Freshness:** small hot HNSW index for recent inserts merged with the big IVF-PQ cold index; nightly compaction.

**Interview takeaway:** The number 500M vectors → 1.5 TB raw is what justifies every later decision (PQ, sharding, rerank). Derive it early and let it drive the design.

## 14 Architecture / Diagrams

### 14.1 Full RAG Pipeline (ingest + query)



## 14.2 HNSW search

QUERY  $q = [0.12, -0.43, 0.87, \dots]$

Layer 2 (sparse, long links): [N1] — [N8]

Layer 1 (medium): [N1] — [N3] — [N5] — [N8]

Layer 0 (all nodes, local): [N1] — [N2] — [N3] — [N4] — [N5] — [N6] — [N7] — [N8] — [N9]

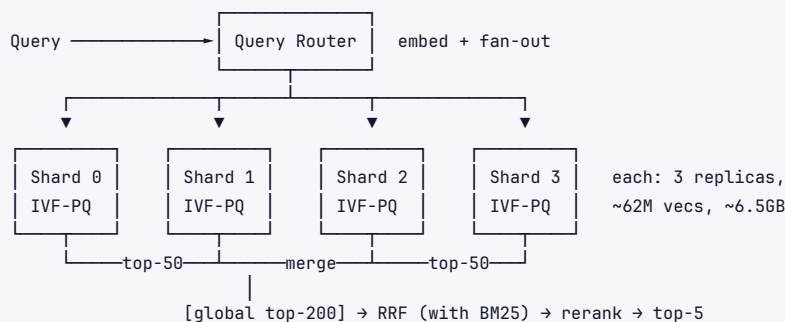
1. Enter top layer at N1, greedily hop toward  $q \rightarrow N8$
  2. Descend to L1, hop  $\rightarrow N5$
  3. Descend to L0, expand candidate set (ef\_search), refine  $\rightarrow$  top-k
- Result: approx nearest neighbors in  $\sim O(\log N)$  hops.

## 14.3 IVF-PQ retrieval

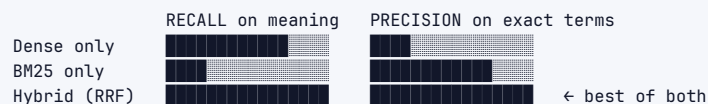
Query  $q$

- ▼ find nprobe nearest centroids (coarse quantizer)  
[C12][C57][C90][...] ← nprobe=64 cells chosen out of nlist=90,000
- ▼ scan only those cells' PQ codes (96 bytes each) via ADC lookup table  
[approx top-200]
- ▼ (optional) rerank with exact float vectors from SSD  
[top-10]

## 14.4 Sharded vector service



## 14.5 Vector vs lexical vs hybrid



# 15 Real-World Examples

## 15.1 Google Search / YouTube (ScaNN)

Embedding-based retrieval ("neural matching") finds semantically related results; ScaNN's anisotropic quantization serves billion-vector MIPS in milliseconds. YouTube recommendations are two-tower retrieval (user tower, video tower)  $\rightarrow$  ANN over the video embedding index. **Lesson:** at Google scale, quantization quality (recall-per-byte) is the whole ballgame.

## 15.2 Meta Ads & FAISS

Ads retrieval: a request embeds user/context, FAISS returns thousands of candidate ads from billions, then a heavy ranker scores them. FAISS IVF-PQ on GPU drives the candidate-generation stage. **Lesson:** two-stage (cheap ANN retrieve → expensive rerank) is universal.

## 15.3 Spotify (Annoy)

Music recommendations: embed tracks/users, Annoy's mmap-able read-only forest serves the same static daily-built index across many machines cheaply. **Lesson:** for a static corpus, an immutable mmap index is operationally simple and memory-efficient.

## 15.4 Pinterest visual search (PinSage)

Embed pins (image + text), ANN retrieve visually/semantically similar pins over billions. **Lesson:** multimodal embeddings + ANN power "more like this."

## 15.5 Perplexity / enterprise RAG (Glean, Notion AI, Microsoft 365 Copilot)

Query → hybrid retrieve (dense + BM25) over chunked corpus → rerank → LLM with citations. Copilot adds **ACL-aware filtered search** — you only retrieve documents you're permitted to see (the pre/post-filter correctness issue is a real security requirement, not just performance). **Lesson:** filtering correctness = security boundary.

## 15.6 Semantic cache for LLMs

Embed each incoming LLM query; if a past query's vector is within cosine 0.97, return the cached answer — skip an expensive LLM call. A vector DB used as a cache. **Lesson:** vector search isn't only for documents.

## 15.7 Near-duplicate detection / dedup

Embed content, find pairs above a high similarity threshold to dedup a corpus or detect plagiarism / repost spam. **Lesson:** the same NN machinery serves dedup, not just search.

# 16 Real-Life Analogies

*One vivid frame: a vast, smart library where books are placed by meaning, not just by title.*

| Concept                             | Library Analogy                                                                                                    |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Embedding</b>                    | A book's location in a "meaning map" — books on similar topics sit on nearby shelves, regardless of title spelling |
| <b>Vector space</b>                 | The whole floor plan where proximity = topical similarity, not alphabetical order                                  |
| <b>Dimensionality (768)</b>         | The number of independent ways books can differ (topic, tone, era, difficulty, ...) — 768 such axes                |
| <b>Cosine similarity</b>            | Comparing the <i>direction</i> you walk to reach two books from the entrance, ignoring how far each is             |
| <b>Normalization (unit vectors)</b> | Putting every book at the same distance from the entrance, so only the <i>direction</i> (meaning) matters          |
| <b>Exact kNN</b>                    | Walking past <i>every</i> shelf to find the closest book — correct but you'd spend all day                         |
| <b>Curse of dimensionality</b>      | In a building with 768 floors, every book ends up "near a wall," so shortcuts (skip whole wings) stop working      |

| Concept                            | Library Analogy                                                                                                                   |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>ANN</b>                         | A clever guide who takes you <i>almost</i> always to the right shelf in seconds, occasionally missing by one                      |
| <b>HNSW</b>                        | Express elevators (top floors, few stops) to get you to the right wing fast, then local stairs to the exact shelf                 |
| <b>IVF (clusters)</b>              | The library is divided into themed rooms; you only enter the few rooms nearest your topic                                         |
| <b>nprobe</b>                      | How many themed rooms you're willing to walk into before settling for what you found                                              |
| <b>Product Quantization</b>        | Replacing each book with a compact index card describing it — 32× lighter to carry, slightly fuzzy                                |
| <b>Reranking (cross-encoder)</b>   | After a runner brings 50 candidate books, the expert librarian reads each against your exact question and hands you the best 5    |
| <b>Hybrid search</b>               | Using both the meaning-map <i>and</i> the old title catalogue, because some patrons ask by exact title (an ISBN) and some by vibe |
| <b>Chunking</b>                    | Cutting fat books into chapters so the runner can hand you just the relevant chapter, not the whole tome                          |
| <b>Metadata filter</b>             | "Only books from the law section that you're cleared to read" — applied while searching, not after                                |
| <b>Tombstone delete</b>            | A "withdrawn" sticker on a book still on the shelf — ignored by patrons until the next big reshelving                             |
| <b>Re-embedding (model change)</b> | The library adopts a new classification system — every book must be re-placed and the whole map redrawn                           |
| <b>Recall@k</b>                    | Of the 10 best books that exist for your question, how many the guide actually brought you                                        |

## 17 Memory Tricks / Mnemonics

### ANN index pick — "HIP-LAS"

- › **HNSW** — default (graph, high recall, high memory)
- › **IVF** — clusters (probe nearby cells)
- › **PQ** — compress (codes, low memory)
- › **LSH** — hashing (similar collide)
- › **Annoy** — trees (mmap, immutable)
- › **ScaNN** — anisotropic (Google, fast MIPS)

**The trade-off triangle — "RLM, pick a point"** Recall / Latency / Memory — you optimize two, sacrifice the third. (Reranking lets you cheat a little.)

**Distance default — "Cosine for text; normalize and it's just dot."** Unit vectors  $\Rightarrow$  cosine = dot product = (monotone with) L2. Same neighbors.

**HNSW knobs — "M builds it, ef searches it."** M and ef\_construction set graph quality at build time; ef\_search tunes recall at query time (the live knob).

**IVF knobs — "nlist cuts the cake, nprobe takes the slices."** More nlist = smaller cells; more nprobe = more cells searched = higher recall.

**RAG retrieval order — "Chunk, Embed, Retrieve, Fuse, Rerank, Generate" → "CERFRG"** ("Can Every Robot Find Really Good [answers]")

**RRF formula — "one over k-plus-rank, summed."**  $\sum 1/(60 + \text{rank})$ .

**Bi vs cross — "Bi is fast and blind; cross is slow and sees."** Bi-encoder embeds separately (scalable retrieval); cross-encoder reads the pair together (accurate rerank).

**The hidden cost — "New model, new map."** Change the embedding model  $\Rightarrow$  re-embed everything  $\Rightarrow$  rebuild the index.

## 18 Common Interview Questions

### 18.1 Q1: Why not just use exact kNN or a k-d tree for vector search?

**Model answer:** Exact kNN is  $O(N \cdot d)$  per query — at 100M vectors  $\times$  768 dims that's ~77B operations, multiple seconds per query, unusable. k-d trees (and other space-partition trees) work in low dimensions but collapse under the curse of dimensionality: in high-d, distances concentrate (nearest and farthest neighbors become nearly equidistant) and almost all volume is near cell boundaries, so the pruning that makes trees fast almost never fires — they degrade to near-brute-force. That's why we use ANN (HNSW/IVF/PQ): accept ~95–99% recall for a 100–1000 $\times$  speedup.

**Follow-ups:** *At what N do you switch from brute force to ANN?* (~100K–1M; below that, exact is fine and simpler.) *What's the recall cost?* (Tunable via `ef_search/nprobe`; measure on a golden set.)

### 18.2 Q2: Explain HNSW. Why is it the default?

**Model answer:** HNSW is a hierarchical, navigable small-world graph — a skip-list idea applied to nearest-neighbor search. Upper layers are sparse with long-range links (express lanes); the bottom layer holds all nodes with short local links. You enter at the top, greedily hop toward the query, descend layer by layer, and refine at layer 0 keeping `ef_search` candidates. Search is  $\sim O(\log N)$ . It's the default because it gives the highest recall at a given latency, needs no training step, supports incremental inserts, and exposes a query-time recall knob (`ef_search`). Its costs are high memory (stores full vectors plus graph edges — ~3.5 KB/vector at  $d=768$ ,  $M=32$ ) and awkward deletes (tombstones + periodic rebuild).

**Follow-ups:** *What do M, ef\_construction, ef\_search do?* (Graph degree; build-time search width; query-time search width.) *Memory for 100M vectors?* (~358 GB.) *How do you delete?* (Tombstone, filter at query, rebuild when bloat exceeds ~25%.)

### 18.3 Q3: How does IVF-PQ let you index a billion vectors?

**Model answer:** Two ideas combined. **IVF** k-means-partitions the space into `nlist` Voronoi cells; at query time you only search the `nprobe` cells nearest the query, skipping ~99% of vectors. **PQ** compresses each vector: split it into `m` sub-vectors, learn a 256-entry codebook per subspace, and store each vector as `m` 1-byte codes — e.g., 768 floats (3072 B)  $\rightarrow$  96 bytes, a 32 $\times$  reduction. At query time, asymmetric distance computation precomputes a lookup table so each candidate's approximate distance is `m` table adds. For  $1B \times 768$ : raw float32 = 3.07 TB (won't fit), PQ = 96 GB (fits). IVF avoids scanning everything; PQ makes what you do scan tiny. Add a rerank of the top candidates with exact float vectors to recover recall.

**Follow-ups:** *Why asymmetric (don't quantize the query)?* (Higher accuracy — query stays full precision.) *Downside of PQ?* (Lossy  $\rightarrow$  lower recall; needs training; retrain on drift.) *How to recover recall?* (Rerank top-N with exact distances.)

### 18.4 Q4: Cosine vs dot product vs L2 --- which and why for text?

**Model answer:** Cosine is the text default: it compares direction (meaning), ignoring magnitude (which correlates with length, not meaning). But if I L2-normalize the vectors — which most text embedding models do — then dot product equals cosine (since  $\|a\|=\|b\|=1 \Rightarrow a \cdot b = \cos$ ), and L2 produces the *same ranking* ( $\|a-b\|^2 = 2 - 2\cos$ ). So in practice I normalize once and configure the index for inner product, getting exact cosine ranking with less per-query compute. L2 is the FAISS default and fine on normalized vectors; un-normalized, cosine and

L2 differ because L2 is magnitude-sensitive.

**Follow-ups:** *When would dot product (not cosine) be right?* (Recsys where magnitude encodes popularity/confidence — MIPS.) *Prove cosine=dot for unit vectors.* ( $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos\theta = \cos\theta$ .)

### 18.5 Q5: How do you choose chunk size, and what's the trade-off?

**Model answer:** Chunks are what you embed and retrieve, so size controls precision vs. context. Too small (e.g., 64 tokens): high precision but the chunk lacks enough context to answer ("Click Settings, then Security" without the rest of the steps). Too large (e.g., 2000 tokens): the relevant sentence is diluted by irrelevant text — the averaged embedding is fuzzy (low precision) and you waste the LLM's context window. The sweet spot is 256–512 tokens with 10–20% overlap so a sentence split at a boundary still appears whole somewhere. For structured docs I split on headers; for high precision I use parent-child (retrieve a small child, feed the larger parent to the LLM).

**Follow-ups:** *What's overlap for?* (Avoid cutting an answer across a boundary.) *Parent-child?* (Embed small chunks for precise retrieval, but pass the surrounding parent to the LLM for context.)

### 18.6 Q6: Walk me through filtered search. What's the gotcha?

**Model answer:** The gotcha is **post-filtering**: if you ANN-retrieve top-k and *then* drop non-matching results, a selective filter (say 1% of vectors match) can leave you with zero results even though good matches exist — a correctness bug — and over-fetching to compensate is slow. Naive **pre-filtering** breaks HNSW because the graph navigates through nodes that may be filtered out, collapsing recall. The production answer is **filtered ANN**: the predicate is fused into graph traversal (Qdrant, Weaviate, Milvus do this), or you partition by the filter key (one index per tenant), or — for highly selective filters — fall back to exact brute force over the tiny matching subset. In ACL-aware enterprise RAG this isn't just performance; it's a security boundary.

**Follow-ups:** *How does partitioning help?* (Each partition is its own small index, queried directly — no recall loss.) *When is brute force fine?* (When the filter leaves few candidates.)

### 18.7 Q7: Why hybrid search, and how do you combine the two rankings?

**Model answer:** Dense embeddings excel at paraphrase/concept matching but miss exact strings — SKUs, error codes, rare proper nouns the model never saw. BM25 nails those exact terms but misses synonyms. Neither alone is best, so I run both and fuse with **Reciprocal Rank Fusion**: score each doc by  $\sum 1/(k+\text{rank})$  across the two lists ( $k \approx 60$ ). RRF uses only ranks, so it sidesteps the problem that BM25 scores and cosine scores aren't comparable. A doc both methods rank highly wins; the method works without score normalization or tuning.

**Follow-ups:** *Why rank-based not score-based fusion?* (Scores live on incomparable scales; ranks are robust.) *What does k control?* (Higher k flattens the contribution of top ranks — smoother fusion.)

### 18.8 Q8: Explain retrieve-then-rerank. Why two stages?

**Model answer:** Retrieval uses a bi-encoder — query and documents embedded independently — which is fast and scalable (embed the corpus once offline, ANN at query time) but coarse, since the query and document never attend to each other. A cross-encoder feeds [query, doc] together through full attention and outputs a precise relevance score, but it needs one forward pass per document — impossible over 100M docs. So: stage 1 retrieves top-30–50 cheaply with the bi-encoder ANN; stage 2 reranks just those 30–50 with the cross-encoder and keeps the top 3–5. You get the cross-encoder's accuracy at the bi-encoder's scale. Reranking ~50 docs adds ~40 ms but materially improves answer quality.

**Follow-ups:** *Why not cross-encode everything?* ( $O(N)$  forward passes — billions of transformer runs.) *Commercial options?* (Cohere/Voyage/Jina rerankers.)

### 18.9 Q9: You upgrade your embedding model. What happens operationally?

**Model answer:** A new model produces a different vector space — old and new vectors aren't comparable — so you must **re-embed the entire corpus and rebuild the index**; you can't mix or incrementally migrate. I'd do blue/green: stand up a new (green) index, re-embed everything into it, validate recall@10 on the golden set against the live (blue) index, dual-write new ingests during transition, then cut over and decommission blue. Budget roughly 28 GPU-hours per 100M chunks plus  $2\times$  storage during overlap. This re-embed cost is the single biggest hidden operational risk of a vector system.

**Follow-ups:** *Can you avoid re-embedding?* (No — geometry changes. Matryoshka embeddings let you truncate dims without re-embedding, but a *new model* still requires it.) *How do you validate the new index?* (Golden-set recall must meet or beat the old.)

### 18.10 Q10: How do you measure whether retrieval is good?

**Model answer:** Build a golden set of (query, relevant\_doc\_ids) — from expert labels, click logs, or LLM-generated questions per chunk. Then track recall@k (did we find the relevant docs?), precision@k (are top-k clean?), MRR (how high is the first hit?), and nDCG@k (graded ranking quality). Crucially, separate **ANN recall** (index returns the true nearest vectors, ground-truthed by brute force) from **retrieval recall** (we return the truly relevant documents, ground-truthed by labels): a bad embedding model gives perfect ANN recall but poor retrieval recall. Gate every index/model/chunking change on offline metrics, then A/B test online for click-through and answer success.

**Follow-ups:** *Define nDCG.* ( $DCG = \sum rel_i / \log_2(i+1)$ , normalized by ideal DCG; rewards ranking the best doc first.) *Recall 0.99 but bad answers — why?* (Embedding model or chunking, not the index.)

### 18.11 Q11: Design semantic search over 100M documents with <100 ms p95.

**Model answer:** 100M docs  $\times$  ~5 chunks  $\approx$  500M vectors. At d=768 float32 that's 1.5 TB raw — too big for one box, so memory drives the design. Use bge-base (768-dim, self-hosted, normalized  $\rightarrow$  inner product), recursive 512/64-token chunks with metadata. Index: IVF-PQ (nlist $\approx$ 90K, nprobe=64, m=96 8-bit codes  $\rightarrow$  ~50 GB codes) plus a rerank of top-200 against exact float vectors on SSD. Shard by hash into 8 shards (~62M each), 3 replicas; scatter-gather and merge. Add BM25, fuse with RRF, then cross-encoder rerank to top-5. Budget: ~5 ms query embed + ~25 ms ANN + ~15 ms BM25 (parallel) + ~10 ms SSD rerank + ~40 ms cross-encoder  $\approx$  80 ms p95, recall@10  $\approx$  0.96, ~150 GB RAM (3 $\times$  replicated). If memory weren't constrained I'd use HNSW (M=32) for ~0.97 recall and skip the rerank, at ~1.75 TB RAM.

**Follow-ups:** *Why IVF-PQ over HNSW here?* (35 $\times$  memory savings; HNSW = 1.75 TB.) *How do filters/freshness work?* (Tenant partition + filtered IVF; hot HNSW index for recent inserts merged with cold IVF-PQ.)

### 18.12 Q12: How do deletes work in a vector index?

**Model answer:** Deletes are the hard operation. HNSW graphs can't cleanly remove a node without orphaning neighbors' links, so the standard approach is a **tombstone**: mark the vector deleted, leave it in the graph, and filter it out of query results. Tombstones accumulate — wasting memory and degrading recall because the graph still routes through dead nodes — so when the tombstone ratio passes ~20–30% you **rebuild** the index from live vectors (often as a shadow index, then swap), analogous to LSM-tree compaction. IVF handles it more gracefully (drop the id from a cell's posting list), though churn eventually unbalances cells and triggers a retrain.

**Follow-ups:** *Why not remove the HNSW node immediately?* (Breaks graph navigability.) *How does this resemble databases?* (Tombstones + compaction = LSM-tree; MVCC dead tuples + VAC-

UUM.)

## 19 Senior-Level Discussion Points

### 19.1 Memory is usually the binding constraint, not CPU

At scale, the question "will the index fit in RAM?" dictates the architecture. HNSW stores full vectors (~3.5 KB/vec at  $d=768$ ) → great recall, but 100M = 358 GB and 1B = 3.5 TB. PQ trades recall for a  $32\times$  memory cut. The senior move is to start from the raw-size estimate ( $N \times d \times 4\text{ B}$ ) and let it choose between HNSW (memory available) and IVF-PQ (+ rerank) (memory bound). Disk-based ANN (DiskANN, Milvus) pushes vectors to NVMe with a small in-RAM graph for even larger scale at higher latency.

### 19.2 Two-stage retrieval is universal

Cheap-recall then expensive-precision appears everywhere: ANN retrieve → cross-encoder rerank; IVF-PQ approx → exact float rerank; candidate generation → heavy ranker (ads/recs). Always retrieve more than you show. The cost asymmetry (cheap stage over millions, expensive stage over dozens) is the whole point.

### 19.3 Embedding quality dominates index quality

Teams obsess over `ef_search` while their embedding model is the actual bottleneck. A perfect index over bad vectors returns the wrong neighbors faithfully. Diagnose by separating ANN recall (vs brute force) from retrieval recall (vs golden labels). Often the highest-ROI move is fine-tuning the embedding model on in-domain (query, doc) pairs, not tuning the index.

### 19.4 Hybrid + rerank beats bigger models

Before reaching for a 3072-dim model or a 7B embedder, add BM25 + RRF + a cross-encoder reranker. Hybrid fixes the exact-term blind spot of dense vectors; reranking fixes ordering. This stack usually beats a larger single bi-encoder at lower cost.

### 19.5 Filtering correctness is a security boundary

In multi-tenant / ACL systems, "retrieve only what this user can see" is not a perf nicety — leaking a retrieved chunk is a data breach. Post-filtering can both miss results (correctness) and, if implemented sloppily, leak. Use filtered-ANN or partition by tenant, and treat the filter as part of the authorization path.

### 19.6 The re-embed treadmill

Every embedding-model upgrade forces a full corpus re-embed + index rebuild. This caps how often you can chase model improvements and argues for (a) self-hosting embeddings to avoid per-call costs at re-embed time, (b) Matryoshka models to get a dim knob without re-embedding, and (c) blue/green pipelines so upgrades are routine, not heroic.

### 19.7 Freshness vs rebuild cost

Real-time use cases (chat memory) need second-level freshness; doc search tolerates hours. The hot/cold split (small frequently-rebuilt index + large nightly-rebuilt index, queried together) is the standard answer. State your freshness SLA explicitly; it changes the design.

## 19.8 Quantization recall recovery

PQ's recall hit is recoverable: retrieve a larger candidate set by PQ approx distance, then re-score with exact float vectors (kept on SSD). This decouples "fit in RAM" (PQ codes) from "final ranking accuracy" (exact rerank) — a clean way to break the recall/memory tension.

## 20 Typical Mistakes Candidates Make

1. **Saying "just use cosine similarity" with no mention of normalization or the dot-product equivalence.** The signal is knowing that normalized vectors make cosine = dot, so you configure the index for inner product.
2. **Proposing exact kNN or a k-d tree at scale.** Shows no awareness of  $O(N \cdot d)$  cost or the curse of dimensionality. Name ANN immediately for  $N > \sim 1M$ .
3. **Treating filtering as a post-step.** Post-filtering causes the empty-results correctness bug and, in ACL systems, security risk. Always raise filtered-ANN or partitioning.
4. **Confusing bi-encoder and cross-encoder.** Bi = independent embeddings for scalable retrieval; cross = joint attention for accurate reranking. Mixing them up reveals shallow RAG understanding.
5. **Ignoring chunking.** Retrieval quality is dominated by chunk size/overlap/strategy. "I'll embed the documents" without chunking is a red flag.
6. **Forgetting the re-embed cost of model upgrades.** Many candidates think you can swap embedding models freely. You can't — it's a full corpus rebuild.
7. **Conflating ANN recall with retrieval recall.** A great index over a bad embedding model still returns wrong results. Separate the two when diagnosing.
8. **"HNSW always" with no memory math.** HNSW is the default, but at billions of vectors its memory (full vectors + graph) is prohibitive; that's when IVF-PQ wins. Quote the numbers.
9. **Skipping hybrid search.** Pure dense search misses exact terms (SKUs, error codes). Mention BM25 + RRF.
10. **No golden set / no evaluation plan.** "I'd tune ef\_search" without saying *against what metric on what data* is hand-waving. Recall@k on a golden set gates everything.
11. **Claiming PQ is lossless.** PQ is lossy compression; it reduces recall. The fix is reranking with exact vectors, not pretending there's no cost.
12. **Forgetting deletes are hard.** Saying "just delete the vector" ignores HNSW's tombstone-and-rebuild reality.

## 21 How This Connects to Other Topics

| Topic                         | Connection                                                                                                                                                                                          |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LLM Applications / RAG</b> | Vector search is the retrieval engine of RAG; chunking, hybrid, rerank all live here                                                                                                                |
| <b>Databases (08)</b>         | ANN index = the B-tree analog for semantic proximity; tombstones + rebuild mirror LSM compaction + VACUUM; filtered search $\approx$ index + predicate; sharding/replication are identical concepts |
| <b>Distributed Systems</b>    | Sharding (scatter-gather vs routed), replication, consistency/freshness, quorum reads for vector shards                                                                                             |
| <b>Machine Learning</b>       | Embedding models, contrastive/InfoNCE training, bi/cross-encoders, two-tower retrieval, CLIP/multimodal                                                                                             |

| Topic                                   | Connection                                                                                                                           |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data Structures &amp; Algorithms</b> | Skip lists (HNSW intuition), k-means (IVF/PQ), graphs (NSW), priority queues (candidate sets), k-d trees and why they fail in high-d |
| <b>System Design</b>                    | Retrieval service design, latency budgets, capacity estimation, hot/cold indexes, caching                                            |
| <b>Information Retrieval</b>            | BM25/TF-IDF, inverted indexes, RRF, nDCG/MRR/precision/recall, evaluation methodology                                                |
| <b>Performance Engineering</b>          | SIMD distance kernels, memory bandwidth limits, quantization, GPU vs CPU ANN, mmap                                                   |
| <b>Security / Privacy</b>               | ACL-aware filtered retrieval as an authorization boundary; on-device embeddings for privacy                                          |

## 22 FAANG Interview Tips

1. **Lead with the size estimate.** "N vectors  $\times$  d dims  $\times$  4 bytes" is the number that drives every architectural choice. Compute it out loud first.
2. **Always name the recall/latency/memory triangle** and which corner you're optimizing, then name the knob (ef\_search / nprobe) you'd tune against a recall@k target.
3. **Default to HNSW, switch to IVF-PQ when memory-bound.** Justify with the memory math (358 GB vs 96 GB), don't just name-drop.
4. **Never say "filter the results."** Say "filtered-ANN or partition by the filter key; exact brute force for selective filters" — and flag the post-filter correctness/security bug.
5. **Propose hybrid + rerank by default.** Dense + BM25 fused with RRF, then a cross-encoder. It beats a bigger model cheaper.
6. **Use the same embedding model for ingest and query**, normalized, inner-product metric. State the cosine=dot equivalence.
7. **Bring up the re-embed cost** whenever model choice or upgrades come up — it's the senior signal.
8. **Separate ANN recall from retrieval recall** when discussing evaluation; tie everything to a golden set.
9. **For Google, mention ScaNN/anisotropic quantization; for Meta, FAISS and two-tower; for Amazon, OpenSearch kNN and cost.** Company-specific knowledge reads as systems maturity.
10. **State trade-offs explicitly.** "PQ saves  $32\times$  memory but loses recall; I recover it by reranking the top-200 with exact float vectors on SSD." Quantified trade-offs win.

## 23 Revision Cheat Sheet

### 23.1 10-Minute Summary

**The problem:** find nearest vectors by *meaning*. Exact kNN is  $O(N \cdot d)$  (seconds at 100M); k-d trees die to the curse of dimensionality. So use **ANN** — ~95–99% recall for 100–1000 $\times$  speedup.

**Embeddings:** dense learned vectors where geometric closeness = semantic similarity. Trained contrastively (InfoNCE, in-batch negatives, SimCSE). Bi-encoder = independent embeddings (scalable retrieval); cross-encoder = joint attention (accurate rerank). Dims 384/768/1536/3072 trade quality vs memory (linear). Normalize  $\rightarrow$  cosine = dot. Model upgrade = full re-embed + rebuild.

**Metrics:** cosine for text (direction = meaning); normalized  $\Rightarrow$  cosine = dot = (monotone) L2 — same neighbors. Pick inner product on normalized vectors.

**Indexes:**

- › **HNSW** — hierarchical small-world graph;  $\sim O(\log N)$ ; knobs  $M$  /  $ef\_construction$  /  $ef\_search$ ; high recall, high memory (full vectors + graph). **Default.**
- › **IVF** — k-means cells; probe  $nprobe$  nearest cells; needs training.
- › **PQ** — split into  $m$  sub-vectors, codebooks,  $m$ -byte codes;  $32\times$  memory cut; lossy; ADC lookup.
- › **IVF-PQ** — billion-scale workhorse (FAISS).
- › **LSH** (hashing), **Annoy** (mmap trees, immutable), **ScaNN** (anisotropic, Google, fastest MIPS).

**Triangle:** Recall / Latency / Memory — pick a point; tune via  $ef\_search/nprobe$ ; rerank to cheat.

**RAG pipeline:** chunk (512/64 tokens, recursive)  $\rightarrow$  embed  $\rightarrow$  upsert  $\rightarrow$  retrieve top-k  $\rightarrow$  metadata filter (filtered-ANN, not post-filter)  $\rightarrow$  hybrid (dense + BM25, fuse with RRF =  $\Sigma 1/(60+rank)$ )  $\rightarrow$  cross-encoder rerank  $\rightarrow$  top 3–5  $\rightarrow$  LLM.

**Ops:** inserts incremental ( $\sim$ query cost); deletes = tombstone + rebuild; shard (hash scatter-gather vs cluster-routed) + replicate; re-embed on model change (blue/green); freshness via hot/cold split.

**Eval:** golden set; recall@k, precision@k, MRR, nDCG; separate ANN recall (vs brute force) from retrieval recall (vs labels); offline-gate, online A/B.

## 23.2 Key Numbers to Know

- › Exact kNN:  $O(N \cdot d)$ .  $100M \times 768 \approx 77B$  ops  $\approx$  seconds/query.
- › HNSW memory:  $\sim 3.5$  KB/vector ( $d=768, M=32$ )  $\rightarrow 100M \approx 358$  GB,  $1B \approx 3.5$  TB.
- › PQ: 768 floats (3072 B)  $\rightarrow$  96 bytes =  $32\times$  cut.  $1B \times 768$ : 3.07 TB float32  $\rightarrow$  96 GB PQ.
- › IVF:  $nlist \approx \sqrt{N} - 4 \cdot \sqrt{N}$ ;  $nprobe$  = recall knob.
- › Chunks: 256–512 tokens, 10–20% overlap.
- › RRF  $k \approx 60$ .
- › Re-embed:  $\sim 28$  GPU-hours / 100M chunks.
- › Typical recall targets: 0.90 (web) to 0.99 (medical/legal).
- › Dims: 384 / 768 / 1024 / 1536 / 3072; memory linear in  $d$ .

## 23.3 Most Important Concepts for Interviews

| Rank | Concept                                                        | Why                                 |
|------|----------------------------------------------------------------|-------------------------------------|
| 1    | Why exact kNN / k-d trees fail (curse of dimensionality)       | Justifies ANN's existence           |
| 2    | HNSW mechanism + $M/ef$ params + memory                        | The default index; asked constantly |
| 3    | IVF-PQ + the memory math                                       | Billion-scale; quantization         |
| 4    | Recall/latency/memory triangle + tuning knobs                  | The core trade-off                  |
| 5    | Cosine vs dot vs L2 + normalization equivalence                | Distance fundamentals               |
| 6    | Chunking size/overlap trade-off                                | Dominates RAG quality               |
| 7    | Hybrid (dense + BM25) + RRF                                    | Beats dense-only, cheap             |
| 8    | Retrieve $\rightarrow$ rerank (bi vs cross encoder)            | Two-stage architecture              |
| 9    | Filtered search pre/post-filter gotcha                         | Correctness + security              |
| 10   | Re-embed cost on model change                                  | Senior operational signal           |
| 11   | Evaluation: recall@k/nDCG, golden set, ANN vs retrieval recall | How you know it works               |

| Rank | Concept                      | Why         |
|------|------------------------------|-------------|
| 12   | Sharding + tombstone deletes | Scaling/ops |

## 23.4 Cheat-Sheet Summary Table

| Question                        | Quick Answer                                            |
|---------------------------------|---------------------------------------------------------|
| Exact kNN cost?                 | $O(N \cdot d)$ — seconds at 100M; use ANN               |
| Why not k-d tree?               | Curse of dimensionality — pruning fails > ~20 dims      |
| Default ANN index?              | HNSW                                                    |
| Billion-scale, RAM-bound?       | IVF-PQ + rerank                                         |
| Query-time recall knob?         | HNSW <code>ef_search</code> , IVF <code>nprobe</code>   |
| Text distance metric?           | Cosine; normalize → use inner product                   |
| Cosine = dot when?              | Unit-normalized vectors                                 |
| PQ memory cut?                  | ~32× (768 floats → 96 bytes)                            |
| Chunk size?                     | 256–512 tokens, 10–20% overlap                          |
| Combine dense + BM25?           | RRF: $\Sigma 1/(60 + \text{rank})$                      |
| Rerank model type?              | Cross-encoder (joint attention)                         |
| Filter correctly?               | Filtered-ANN or partition; never naive post-filter      |
| Delete from HNSW?               | Tombstone + periodic rebuild                            |
| Model upgrade?                  | Full re-embed + rebuild (blue/green)                    |
| Retrieval metric?               | Recall@k, nDCG, MRR on a golden set                     |
| ANN recall vs retrieval recall? | Index correctness vs system correctness — separate them |
| Spotify's library?              | Annoy (mmap, immutable)                                 |
| Meta's library?                 | FAISS                                                   |
| Google's library?               | ScaNN (anisotropic quantization)                        |